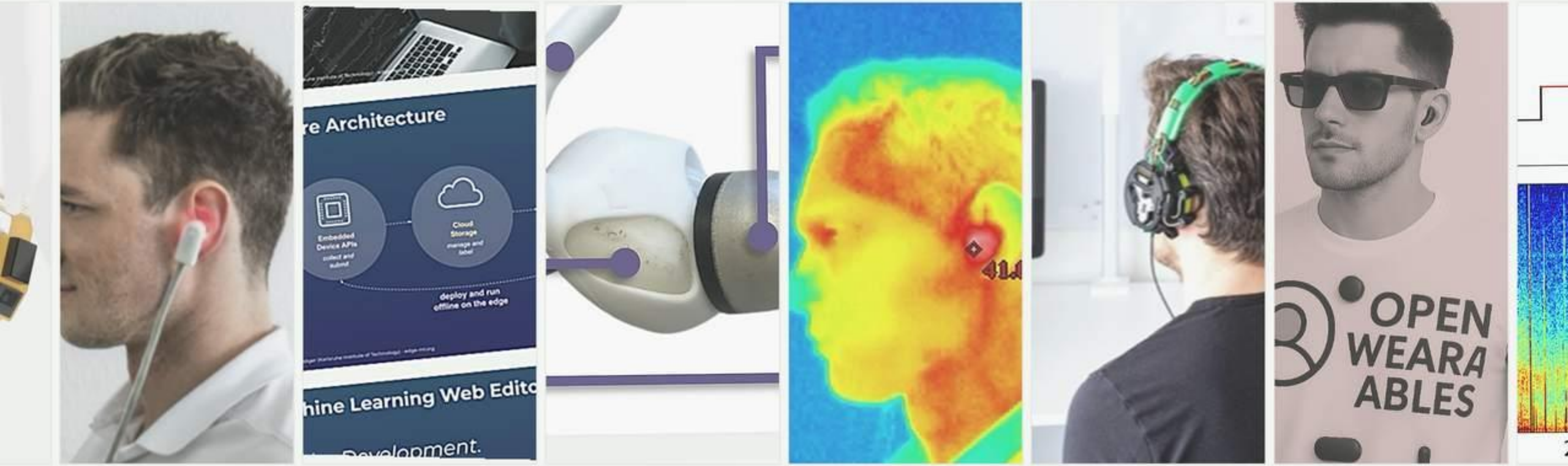


Human Computer Interaction SS26

Chapter 4

Design Analysis

Michael Beigl, TECO

Lecture Overview

Date / Day	Topic
21 Apr 2026 (Tuesday)	Introduction and Overview
22 Apr 2026 (Wednesday)	HIP
28 Apr 2026 (Tuesday)	HIP
29 Apr 2026 (Wednesday)	Exercise HIP / Lecture Integrated
05 May 2026 (Tuesday)	Perception
06 May 2026 (Wednesday)	Perception
12 May 2026 (Tuesday)	Design Analysis
13 May 2026 (Wednesday)	Exercise Design Analysis and Perception
19 May 2026 (Tuesday)	Design Analysis
20 May 2026 (Wednesday)	Design
26 May 2026 (Tuesday)	Pentecost / no lecture
27 May 2026 (Wednesday)	Pentecost / no lecture
02 Jun 2026 (Tuesday)	no lecture
03 Jun 2026 (Wednesday)	Exercise Design Analysis
09 Jun 2026 (Tuesday)	Design
10 Jun 2026 (Wednesday)	Interface
16 Jun 2026 (Tuesday)	Interface
17 Jun 2026 (Wednesday)	Exercise Design and Interface
23 Jun 2026 (Tuesday)	Observation
24 Jun 2026 (Wednesday)	Observation
30 Jun 2026 (Tuesday)	Exercise Observation

Analysis of Design

We already heard about the foundations for Design Analysis in HIP and Perception

This alone will enable us to perform an initial Analysis of a HCI design

This chapter will present specific techniques that go beyond

- Laws that can be applied to calculate timings
- Concepts of understanding that go beyond what we heard in HIP and Gestalt
- Analysis methods that let us found tricky problems
- Analysis tools that allow us to calculate usage time

Foundation: Physical Models & Law →

Laws:
Physical Models &
Fitts' Law

Physical Models

- Laws can help us to estimate certain parameters for our user interface
- Laws are idealized and only valid under very restricted conditions
- Laws may guide us in design

Fitts' law

- Fitts' law can be used to determine the size and location of a screen object
- Physical models can predict efficiency based on the physical aspects of a design
- They calculate the time it takes to perform actions such as targeting a screen object and clicking on it
- Fitts' law states that the time it takes to hit a target is a function of the size of the target and the distance to that target

Fitts' law

Three parts:

- Movement Time (MT) — Quantifies the time it takes to complete a task based on the difficulty of the task (ID) and two empirically derived coefficients that are sensitive to the specific experimental conditions
- Index of Difficulty (ID) — Quantifies the difficulty of a task based on width and distance
- Index of Performance (IP) [also called throughput (TP)] — Based on the relationship between the time it takes to perform a task and the relative difficulty of the task

Fitts' law – Predicting Movement Time (MT)

Original: $MT = a + b \ \log_2(2\,A/W)$

- Components:
  - A = amplitude
  - W = width
- a, b = constants dependent on the input device
- Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. Journal of Experimental Psychology, 47, 381-391.

Improved: $MT = a + b \ \log_2(A/W + 1)$

- improvement of the original Fitts' law
- This still assumes linear movement speed

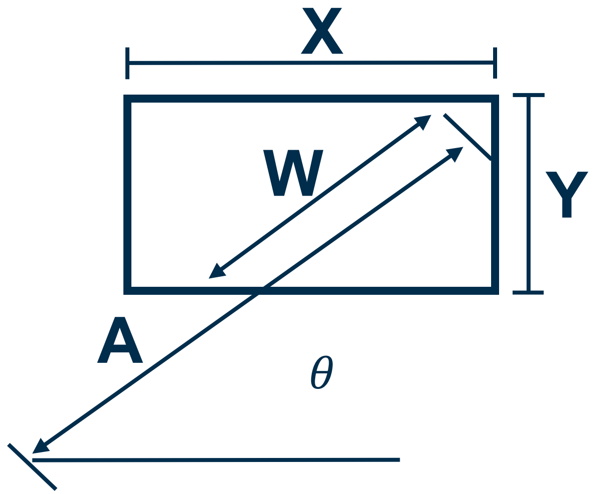
http://www.billbuxton.com/fitts91.html

Further reading: MacKenzie, I. S. (1989). A note on the information-theoretic basis for Fitts' law. *Journal of Motor Behavior*, *21*, 323-330.

Fitts' law – Index of Difficulty (ID)

$$MT = a + b \; log_2(A/W \; + 1)$$

- Components:
  - $\boldsymbol{A}$= distance from starting position
  - $\boldsymbol{W}$= size of target along line of motion (for a 2-D target use):
    - Either smaller of height or depth (bi-variant)
    - The apparent width when approaching (the object)
  - Common values $\mathbf{a} = \mathbf{50}\ \boldsymbol{ms}$, $\boldsymbol{b} = \mathbf{150}\ \boldsymbol{ms/bit}$ → mouse

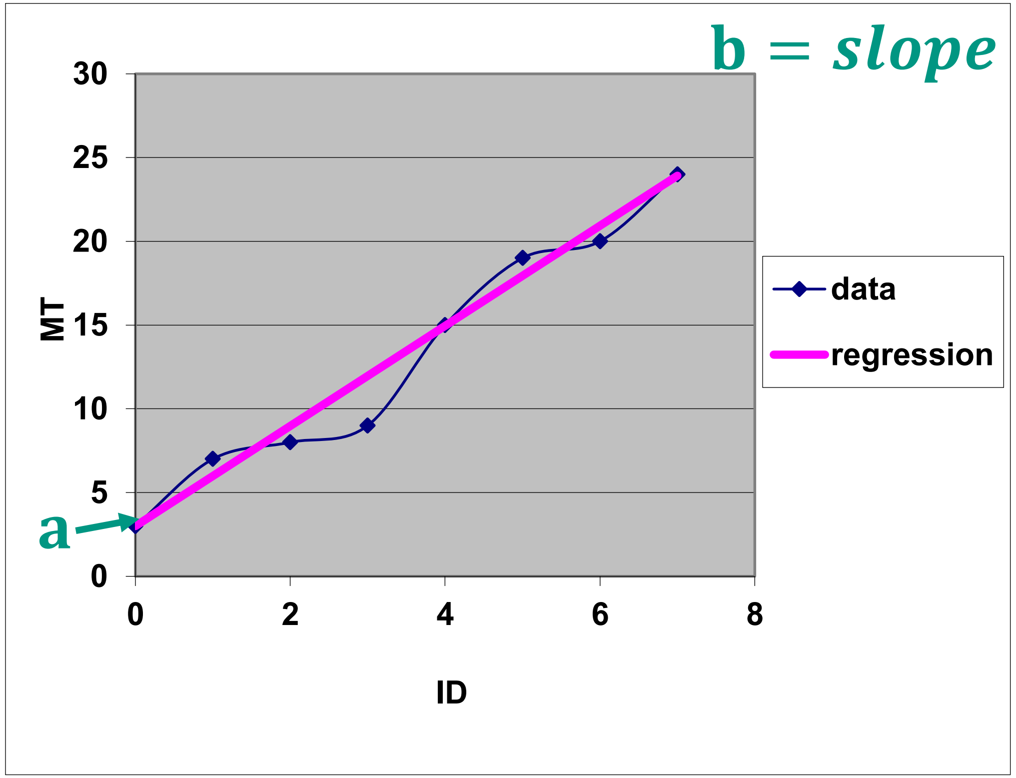


Further reading: Jef Raskin, The Humane Interface, ACM Press 2000, p93-94

Fitts' law – Index of Difficulty (ID)

- how difficult the motor pointing task is
- ID= Index of Difficulty
- $ID = log_2(A/W + 1)$
- ID has the unit bits

- $MT = a + b\ ID$
- $\mathbf{a}$ has the unit s
- $\mathbf{b}$ has the unit s/bits
- Collect data set and
- calculate $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}$ can be negative



linear regression model

Fitts' law – MT difference between task types

- predictive model of time to point at an object
- the time for hand movements is dependent on the distance to move = Amplitude ($\boldsymbol{A}$) and the target size ($\boldsymbol{W}$)
  - doubling the distance increases the time, but does not double it
  - increasing the target size enables more rapid pointing

Movement Time (MT) for "gross-movement tasks":

$$\boldsymbol{MT = a + b\ log_2(A/W + 1)}$$

- where $\boldsymbol{a}$= time to start and stop in seconds for a device
- and $\boldsymbol{b}$= inherent speed of the device e.g. a mouse

Movement Time (MT) for "precision pointing tasks":

$$\boldsymbol{PPMT = a + b\ log_2(A/W + 1) + c\ log_2(d/W)}$$

- where $\boldsymbol{c}$= added constant, dependent on the user's context
- and $\boldsymbol{d}$= distance between hand location and spot where the user first approached the screen

Experimental data for pointing devices

Regression Coefficients

Device	r^a	Intercept, a (ms)	Slope, b (ms/bit>	IP (bits/s)[b]
*** Pointing ***				
Mouse	.990	-107	223	4.5
Tablet	.988	-55	204	4.9
Trackball	.981	75	300	3.3
*** Dragging ***				
Mouse	.992	135	249	4.0
Tablet	.992	-27	276	3.6
Trackball	.923	-349	688	1.5

[a] $n = 16$, $p < .001$

[b] IP (index of performance) = $1/b$

$$MT = a + b\ ID, \quad where\ ID = \log_2(A/W + 1)$$

From http://www.billbuxton.com/fitts91.html

MacKenzie, I. S., Sellen, A., & Buxton, W. (1991). A comparison of input devices in elemental pointing and dragging tasks. Proceedings of the CHI `91 Conference on Human Factors in Computing Systems, pp. 161-166. New York: ACM.

Fitts' law

Implications for interaction design in GUIs with Mouse Interface:

- Overly elongated objects hold no advantage (W/H ratios of $\mathbf{3}$ and higher).
- Objects should be elongated along the most common trajectory path (widgets normally approached from the side should use W, those approached from the bottom or top should use $\mathbf{H}$).
- Objects should not be offset from the screen edge (consistent with the Macintosh OS, not for touch).
- Original Fitts law does not address touch specific problems, e.g. fat finger

(Accot & Zhai, 2003)

Fitts' law

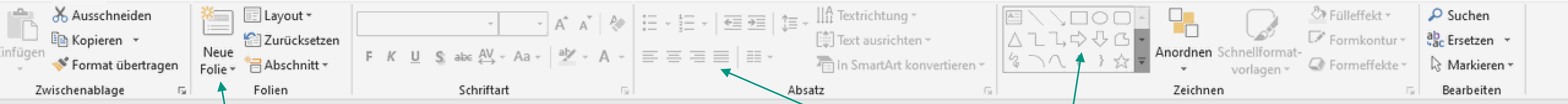
Implications for interaction design in GUIs with Mouse Interface toward touch interfaces

- Are not 1:1 applicable to touch interfaces (and all others) – you have to calculate again
- Fat finger problem
- Fingers are not moving directly from target to target but make a 3D movement
- Fingers may directly reach target or need to be stretched or bended
  - Or even a user may be forced to change to 2-handed input
- Screen size and hand size have a major impact on performance
- But Fitts' law is still good as an initial estimation also for touch interfaces!

(Accot & Zhai, 2003)

Fitts‘ Law

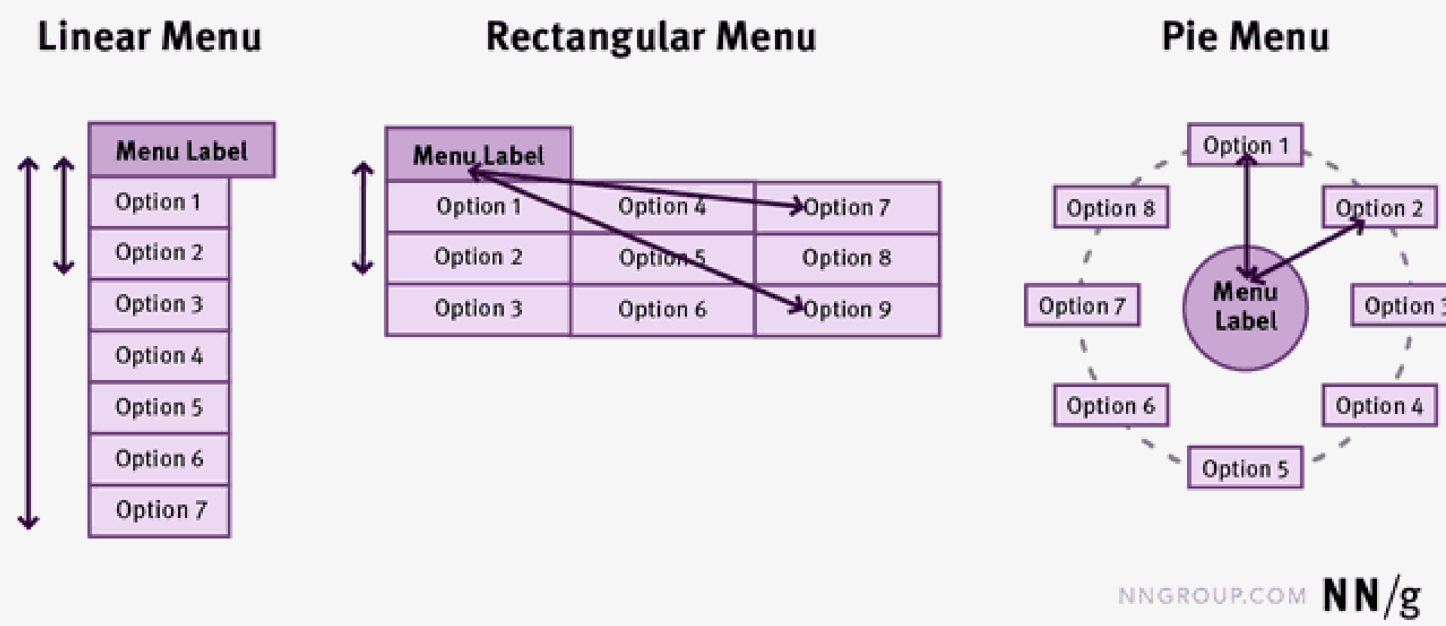
Difficult to access for a pointer when comming from „below“



Simpler to Access

Same Difficulty level as menu, but
More condensed. Saves one click
compared to previous Office versions

Fitts' Law for Menues



The average distance from menu handle to a menu element depends on the type of menu: linear menus are less efficient than rectangular menus, which, in turn, are less efficient than pie menus.

Laws:
Modelling Structure
Hick’s LAW

Hick's law

- The time T needed to make a decision (e.g. a selection) is proportional to the log number of alternatives given
- $\boldsymbol{H}$ is the information-theoretic entropy of a decision
- $\boldsymbol{T=a+b\ H}$
- $\boldsymbol{n}$ alternatives of equal
- probability $\mathbf{H}= \ \boldsymbol{log_2(n+1)}$
- Alternatives of unequal probability
  $\boldsymbol{p_i}$ = the probability of alternative $\boldsymbol{i}$
- $\mathbf{H}=\boldsymbol{\Sigma\, p_i\, log_2(1/\, p_i + 1)}$

Note

Hick's law does not apply if it requires linear search (e.g. a randomly ordered list of commands in a menu). It applies if the user can search by sub-division

http://www.usabilityfirst.com

Hick's law - Modeling structure

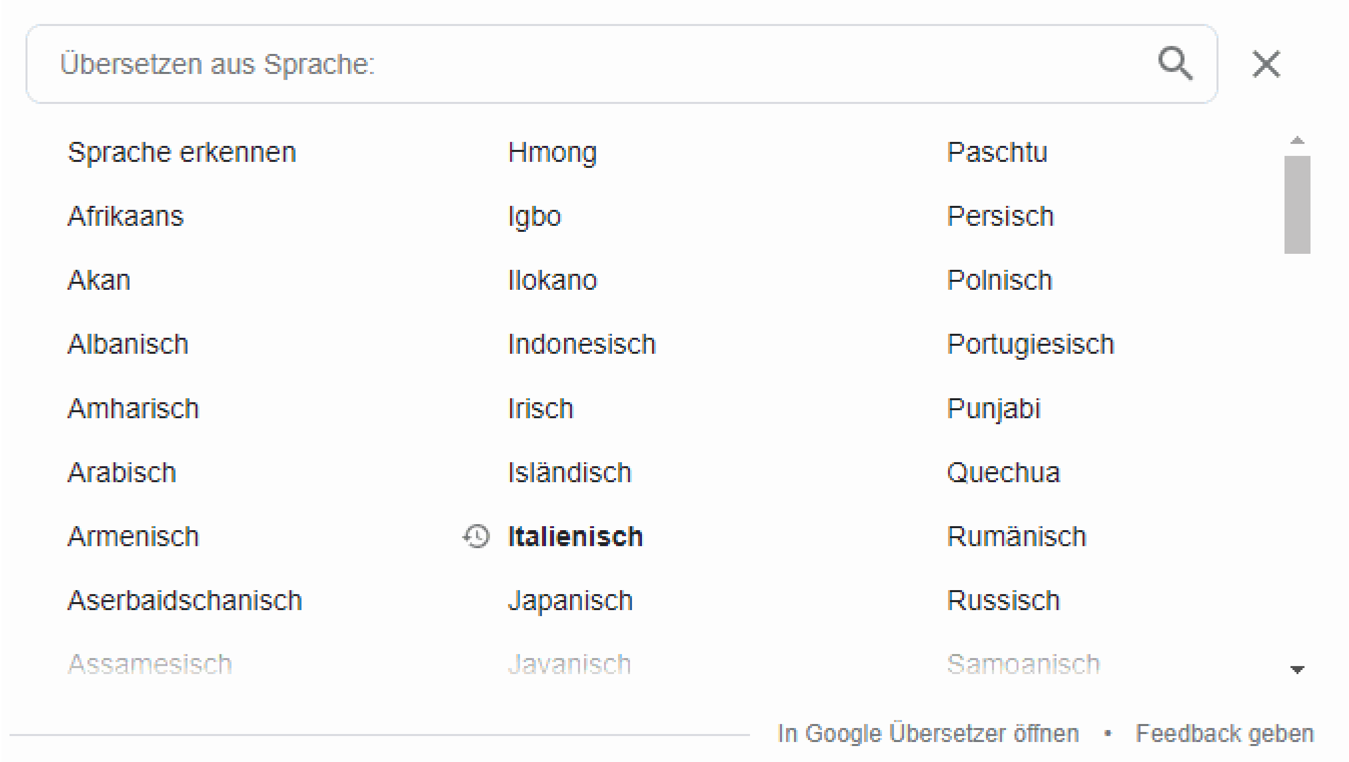
$$T = a + b \log_2(n+1)$$

- Total Decision time in practice (n= number of choices)
- The coefficients are empirically determined from experimental design
- Raskin (2000) suggests that $a = 50\ ms$ and $b = 150\ ms/bit$ are sufficient place holders for "back-of-the-envelope" approximations

- Remember: no linear search, no power law of practice, equal probability

Application of Hicks law: Selection of a long list

Here combined with Fitts‘ law



Google.com

Laws:
Power Law of Practice

Power Law of Practice

- Users get better every time they use a system
  - Neglecting disturbance, decay effects
  - $\boldsymbol{Time = A + B\ N^{-\alpha}}$
- Where:
  - $\boldsymbol{N}$= Trial Number
  - $\mathbf{B}$, $\boldsymbol{\alpha}$= constants (e.g. $\boldsymbol{\alpha}$= .4)
- Strongly dependent on the type of task
  - Motor
  - Cognitive
  - Perception
- … and user

Data Set	Power Law $T = BN^{-\alpha}$ B	α	r^2
Snoddy (1926)	79.20	.26	.981
Crossman (1959)	170.1	.21	.979
Kolers (1975) - Subject HA	14.85	.44	.931
Neisser et al. (1963)			
Ten targets	1.61	.81	.973
One target	.68	.51	.944
Card, English & Burr (1978)			
Stepping keys - Subj. 14	4.95	.08	.335
Mouse - Subj. 14	3.02	.13	.398
Seibel (1963) - Subject JK	12.33	.32	.991
Anderson (Note 1) - Fan 1	2.358	.19	.927
Moran (1980)			
Total time	30.27	.08	.839
Method time	19.59	.06	.882
Neves & Anderson (1980)			
Total time - Subject D	991.2	.51	.780
The Game of Stair			
Won games	1763	.21	.849
Lost Games	980	.18	.842
Hirsch (1952)	10.01	.32	.932

Table 1: Power Law Parameters for the (Log-Log) Linear Data Segments.

Newell, Rosenbloom: Mechanisms of skill acquisition and the law of practice

Concepts:
Mental Models

Mental Models

Mental models “fuse” what we have heard in HIP in perception chapters

People use a mental model to have a basic understanding of what is going on

- The mental model is often unsharp
- E.g., you know your car speeds up when pressing the gas pedal, but the concept behind is unsharp
- Unsharp models might be sufficient, but in certain situations (e.g. exceptions) they might not be enough to explain the system

Mental modeling is method to understand interaction that is both predictive and explanatory

Mental Models and Mapping

Mental models are
- Unscientific: they are based on guesswork
- Partial: They do not describe the whole system
- Unstable: They evolve an adapt to context and experience
- Inconsistent: Some parts of the model may be incompatible with other parts of the system
- Personal: They are specific to individuals

How to come to mental models? Mapping:
- Can be used to foster understanding and building a correct model
- Thus increase usability

Principles of design

Two major principles of design:

Mapping

– To a form and pattern

– To provide overview

Constraints

– To guide

– For support: „affordance“

Applied Design Guidelines & Principles: Mapping

E.g. Error Avoidance Design Guidelines:

Exploit natural mappings

– between intentions and possible actions,

– between actions and their effects on the system state and what is perceivable, and

– between the system state and the needs, intentions and expectations of the user

Principles of good design (Don Norman):

– interface should include good mappings that show the relationship **between stages**

Designing the user interface, B. Shneiderman and C. Plaisant, 4th edition, Addison-Wesley, 2004, chapter 20

Mapping of state and user needs

Most likely user intention?

Applied Design Guidelines & Principles: Mapping

correlation between control element and action

properties for “good” mapping:

- **understandable**
- **consistent**
- **recognizable** or **quickly learnable**
- **natural** (consistent to the user’s knowledge, world and domain knowledge)

example: cooking plates (next slide)

- problems:
  - when written on a button: can be invisible
  - when written next to it: to which button does the text belong?

Designing the user interface, B. Shneiderman and C. Plaisant, 4th edition, Addison-Wesley, 2004, chapter 20

Applied Design Guidelines & Principles: Mapping

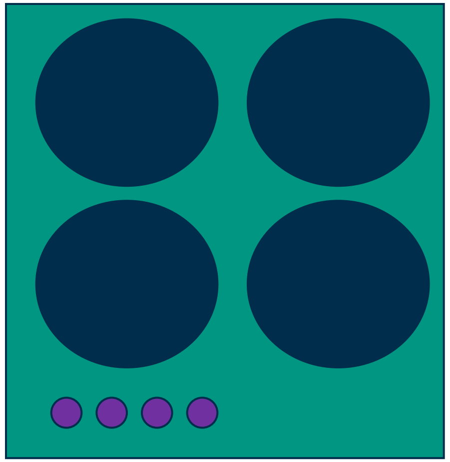
Relationship between controls and action

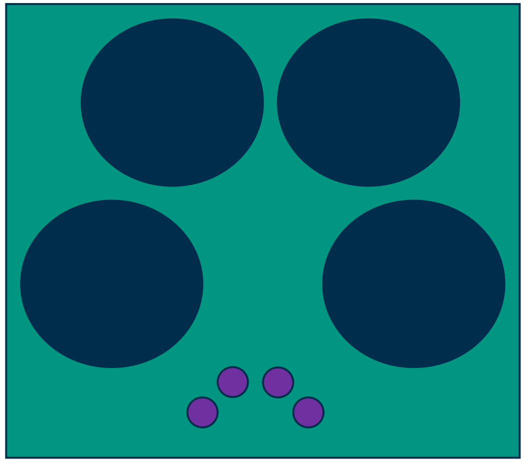
Mappings should be

- Understandable (e.g. moving the mouse up move the slider up)
- Consistent
- Recognizable or at least quickly learnable and easy to recall
- Natural, meaning to be consistent with knowledge the user already has

Example: cooker

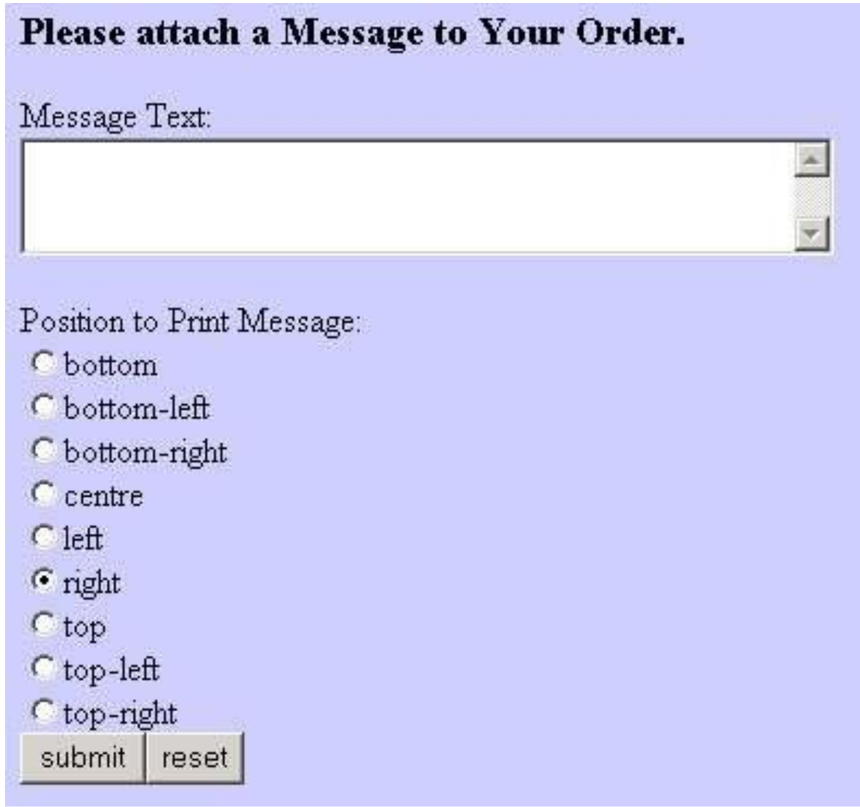
(for these issues see also Gestalt theory)





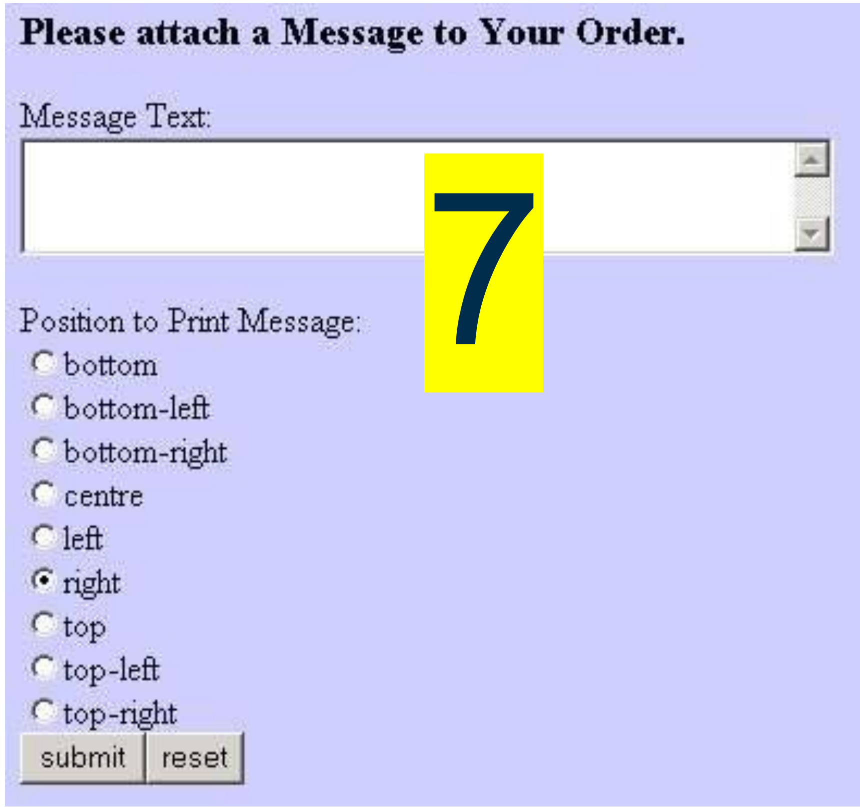
Applied Design Guidelines & Principles: Mapping - Example

Relationship between controls and action



Relationship between controls and action

You can calculate the result by using Tullis complexity calculation



Applied Design Guidelines & Principles: Constraints

- Constrains lead human to build correct mental models
- Constraints guide the user to the next appropriate action or decision
- Constraints also minimize the chance to make errors

Types:

- physical constraints (dial vs. button)
- semantic constraints (assumption that create something meaningful)
- cultural constraints (borders provided by cultural conventions, e.g. traffic signs, colours, ..)
- logical constraints (restrictions due to reasoning)

Applied Design Guidelines & Principles: Constraints – Example 1

https://www.swoodoo.com/, graphically changed

Applied Design Guidelines & Principles: Constraints – Example 2

USB Memory Stick
vs. DVD
- If there is more than one option (physically) cater these cases


▶ image source


▶ image source

Dials vs. Buttons vs. Sliders
- Dials are turned
- Buttons are pressed
- Sliders are pushed


▶ image source

Applied Design Guidelines & Principles: Constraints

- Redundancy is safe
- Constraints can only work at their own level

→ redundant: colors outlet - plug!

→ constraints: square plug – round plug

Designing the user interface, B. Shneiderman and C. Plaisant, 4th edition, Addison-Wesley, 2004, chapter 20

Pingo

Question:
Answer question about “constraints”

Link:
https://pingo.coactum.de/589809

Mental Models: Affordance Concept

Affordance

To build a mental concept of a system

... We need to interpret symbols and components

– This can be seen as 1) matching of the *functionality of the device* and *what we actually want to do*

– This can be seen as 2) the Semantic and Articulator Distance

– Don Norman developed this concept further, and picked up a concept from J.J. Gibson for this: **Affordance**

Affordance – Perception Concept

Don Norman

Human-centered design:

“The Design of Everyday Things”, 1988

affordance refers to the perceived and actual properties of the thing

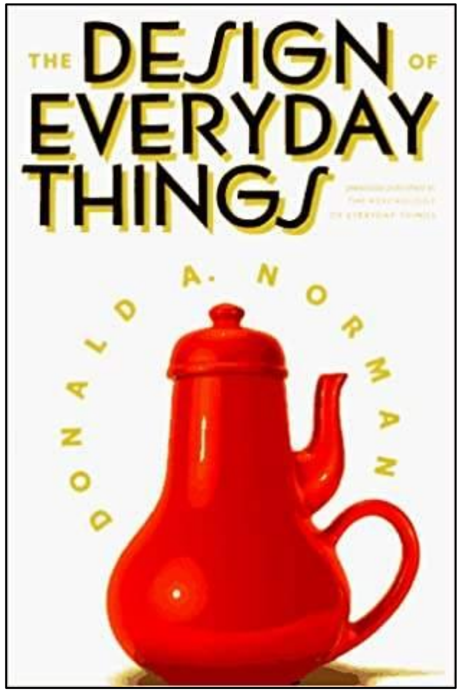
 Match perceived and actual properties

 3 Principles

Make usable properties visible

Use natural associations

Give Feedback



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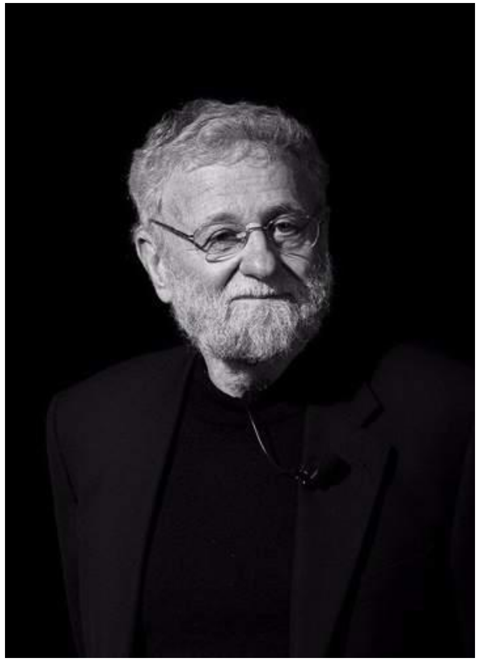

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Affordance - Examples

Objects and environments imply their usage through Gestalt

Different door handels

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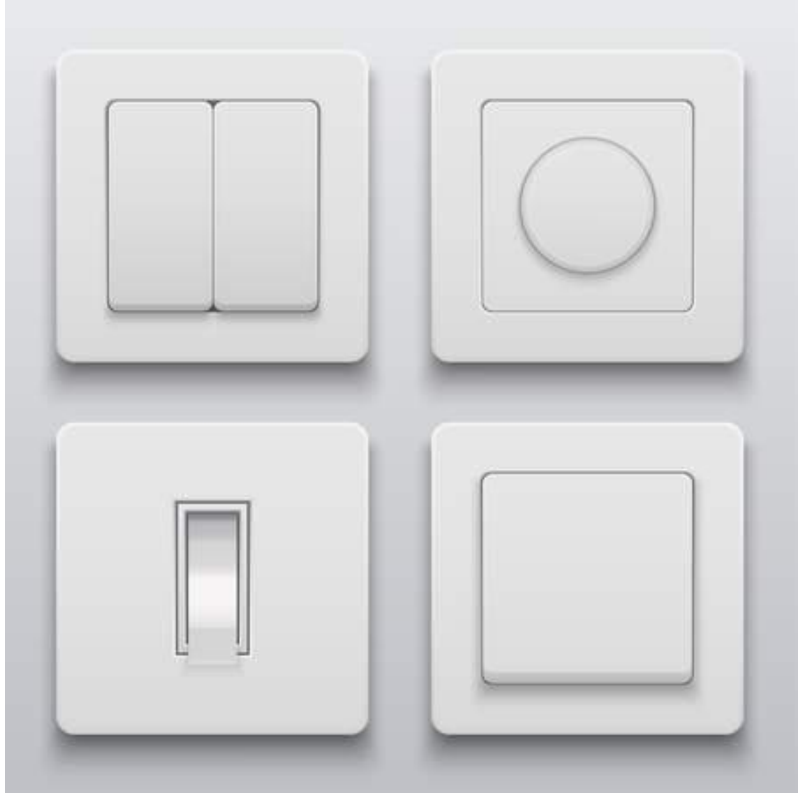
Affordance - Examples

Objects and environments imply their usage through Gestalt

Different light switches



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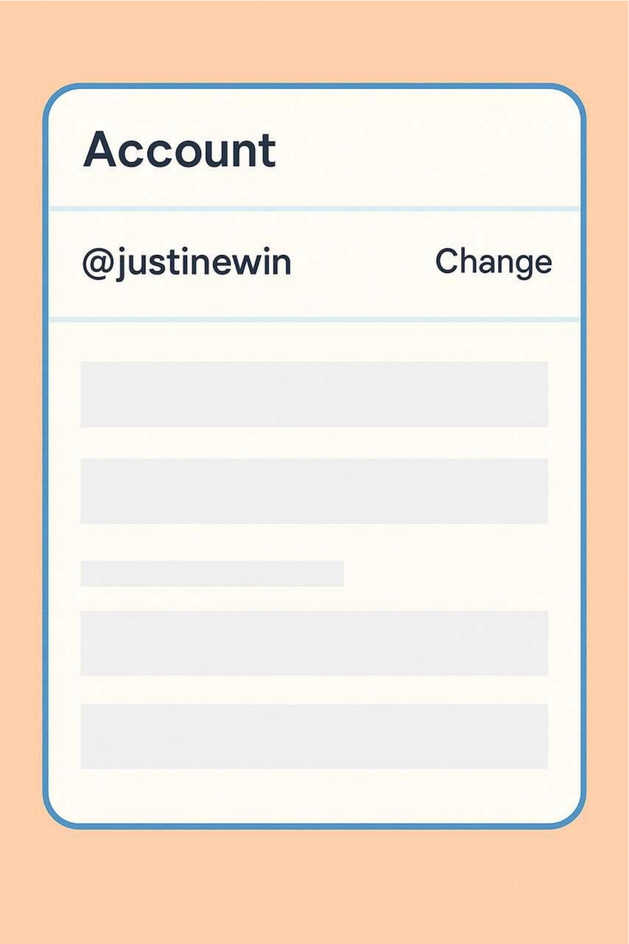
Affordance makes a different

Buttons and Text look differently

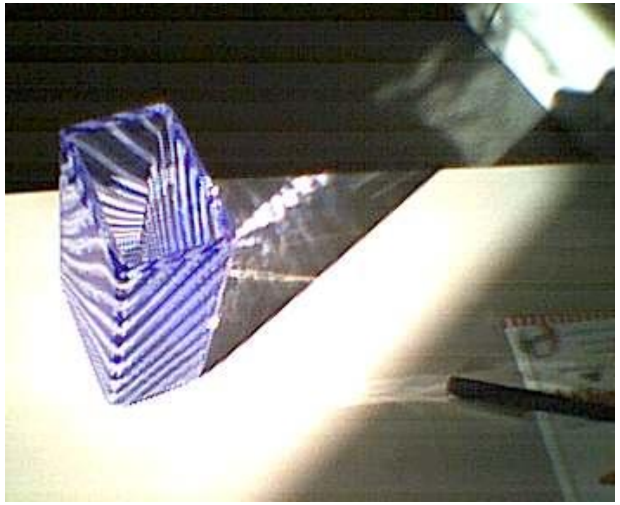
But not always:

– Where is the button in the example on the right side?



Affordance - Examples

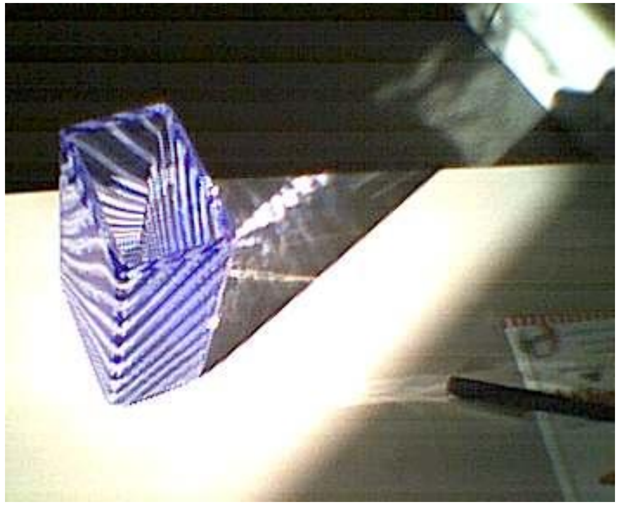
Objects and environments imply their usage through Gestalt



What is this item
used for?

Affordance - Examples

Objects and environments imply their usage through Gestalt



What is this item used for?

Affordance

Question:

- Background information:
- The glass wall of a bus stop is kicked in and has
- to be replaced by the building authority.
- Unfortunately, the destruction is repeated within a
- week, and for reasons of cost, the building
- authority subsequently replaces the glass pane
- with a wood panel.
- Question:
- What happens after that?

Analysis methods →

Analysis of interaction: Mapping Concept applied →

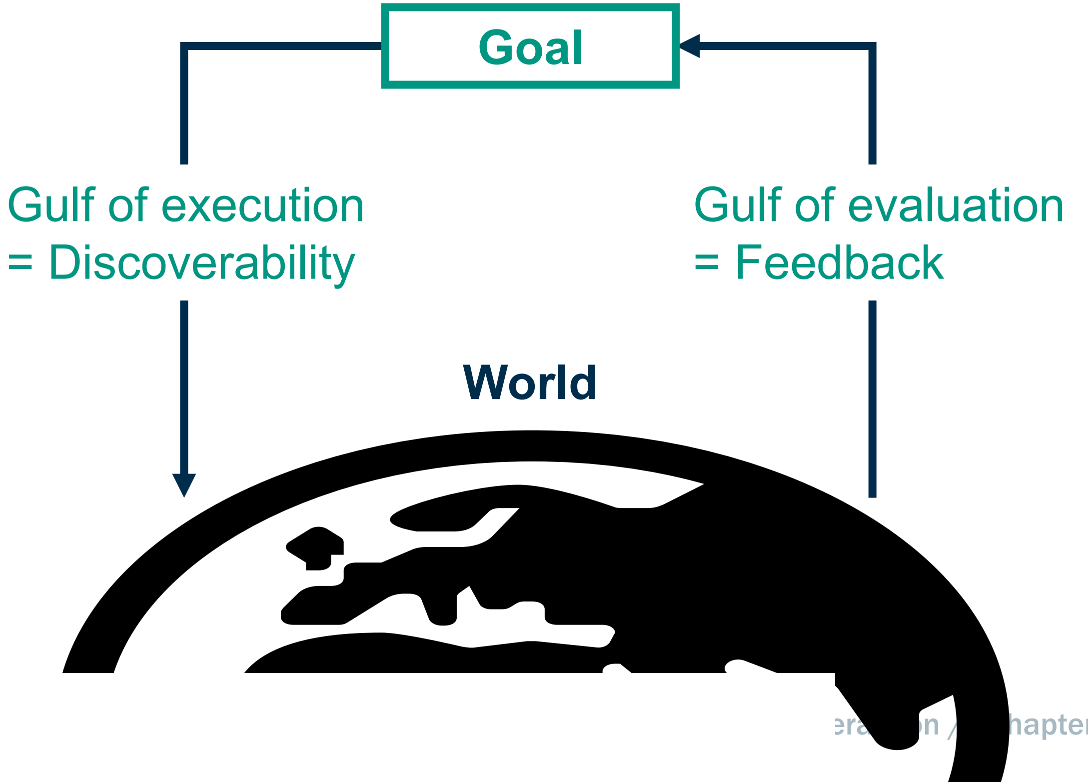
Mapping – What is it good for in Practice

Finding Usability Problems by Evaluate Mapping Strength:

Normans Evaluation/Execution Action Cycle

Execution / Evaluation Action Cycle (EEAC)

– EEAC is a method to look for misunderstandings or difficulties to understand concepts by a user
– The basic concept behind EEAC is to look for two „gulfs“ of understanding: Execution and Evaluation



Execution / Evaluation Action Cycle (EEAC)

- By Don Norman

Goals
We begin with some idea what we want to happen

Execution
We must then execute and act

Evaluation
Finally, validation our action
compare results with goals

World/Action
Manipulate Objects in World

Execution / Evaluation Action Cycle (EEAC)

- 7 stages

Goals
Establish the goal

Execution
Forming intention
Specifying action
Executing action

Evaluation
Evaluating Interpretation
Interpreting Perception
Perceiving World State

World/Action
Manipulate Objects in World

Execution / Evaluation Action Cycle (EEAC)

- Example: Answering machine is full

Goals

Establish the goal: New messages should be recorded

Execution

Forming intention: Delete old messages
Specifying action: press Delete button
Executing action: Look for button on & press

Evaluation

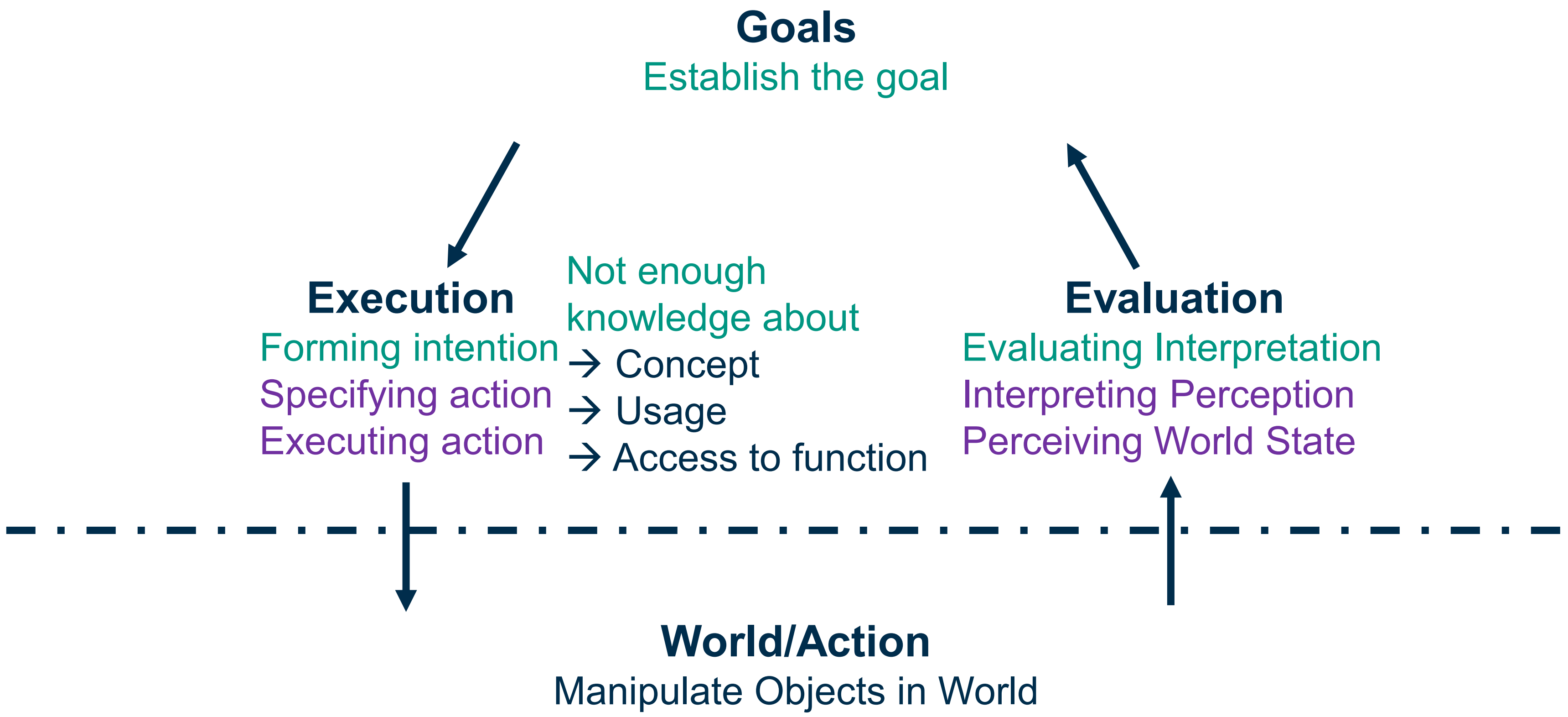
Evaluating Interpretation: Enough place for new messages
Interpreting Perception: no more messages
Perceiving World State: counter shows 0

World/Action

Manipulate Objects in World

Execution / Evaluation Action Cycle (EEAC)

- Gulf of execution: Mismatch between the user's intentions and the allowable actions



Execution / Evaluation Action Cycle (EEAC)

- Gulf of execution: Mismatch between the user's intentions and the allowable actions

Goals

0 Establish the goal

Execution

1 Forming intention
2 Specifying action
3 Executing action

Not enough knowledge about
→ Concept
→ Usage
→ Access to function

Evaluation

8 Evaluating Interpretation
6,7 Interpreting Perception
5 Perceiving World State

Problems with
→ comparing goal and state
→ Interpretation of state
→ perception of state

World/Action

4 Manipulate Objects in World

Execution / Evaluation Action Cycle (EEAC)

- Gulf of evaluation: Mismatch between the system's representation and the users' expectations

Goals
Establish the goal

Execution
Forming intention
Specifying action
Executing action

Evaluation
Evaluating Interpretation
Interpreting Perception
Perceiving World State

Problems with
→ comparing goal and state
→ Interpretation of state
→ perception of state

World/Action
Manipulate Objects in World

Normans Gulf of Execution & Evaluation

Seven Questions to Cross the Gulfs	Goal	Execution	Evaluation	World
What do I want to accomplish?				
What are my alternatives?				
What can I do now?				
How do I do it?				
What happened?				
What does it mean?				
Is this, OK? Have I accomplished my goal?				

Stages of Action Models: Gulf of Execution

The difference between the intentions and the allowable actions is the Gulf of Execution

- How directly can the actions be accomplished?
- Do the actions that can be taken in the system match the actions intended by the person?

Example in GUI

- The user wants a document written on the system in paper (the goal)
- What actions are permitted by the system to achieve this goal?

Good design minimizes the Gulf of Execution

Stages of Action Models: Gulf of Evaluation

The Gulf of Evaluation reflects the amount of effort needed to interpret the state of the system how well this can be compared to the intentions

- Is the information about state of the system easily accessible?
- Is it represented to ease matching with intensions?

Example in GUI

- The user wants a document written on the system in paper (the goal)
- Is the process observable? Are intermediate steps visible?

Good design minimizes the Gulf of Evaluation

Stages of Action Models: Implications on Design

Principles of good design (Norman)

- Stage and action alternatives should be always visible
- Good conceptual model with a consistent system image
- Interface should include good mappings that show the relationship between stages
- Continuous feedback to the user

Critical points/failures

- Inadequate goal formed by the user
- User does not find the correct interface / interaction object
- User many not be able to specify / execute the desired action
- Inappropriate / mismatching feedback

Pingo

Question:
Answer question about “EEAC”
Link:
https://pingo.coactum.de/589809

Task Analysis →

Task Analysis

Definition

Task analysis is the study of how work is achieved by tasks.

- Understanding people and how they carry out their work
- set of methods, not single way
- identify if they reach the goal
- Calculate how long it takes to reach the goal

Benyon et al, Designing Interactive systems, chapter 20

Task Analysis - Definitions

Definition **task**:

A task is a goal together with some ordered set of actions.

definition **goal**:

A goal is a state of the application domain that a work system wishes to achieve.

Goals are specified at particular levels of abstraction.

Benyon et al, Designing Interactive systems, chapter 20

Task Analysis - Definitions

Definition task elaborated:
A task is a structured set of activities required, used or believed to be necessary by an agent to achieve a goal using a particular technology. A task will often consist of subtasks where a subtask is a task at a more detailed level of abstraction. The structure of an activity may include selecting between alternative **actions**, performing some actions a number of times and sequencing of actions

Definition action:
An action is a task which has no problem solving associated with it and which does not include any control structure. Actions and task will be different for different people.

Benyon et al, Designing Interactive systems, chapter 20

Task Analysis - Example

Technology options for achieving the goal: **“join a video meeting”**

- Open Zoom and click the “Join” button, then enter the meeting ID
- Click a Google Calendar event link to launch Google Meet directly
- Ask a smart speaker (e.g., "Hey Google, join my next meeting")
- Use a smartwatch notification to join the meeting with one tap

Issues: pros and cons of each “technology”

- misunderstand may result in suboptimal path
- selection of technology based on available knowledge and access
- after technology selection: tasks can be defined

Benyon et al, Designing Interactive systems, chapter 20

Task Analysis - Categories

Two main categories of task analysis methods:

concerned with the **logic of tasks**

→ sequence of steps to achieve a goal

concerned with the **cognitive aspects**

→ understanding which cognitive processes the work system will have to undertake to achieve a goal

Cognition:

- thinking, solving problems, learning, memory → HIP!
- representations of things that people are assumed to have in their heads: **mental models**

Benyon et al, Designing Interactive systems, chapter 20

Task Analysis – Models

2+1 models in this lecture:

hierarchical task analysis: **HTA**
– Concerned with the logic of the task
goal-based task analysis: **GOMS**
– Concerned with the cognitive analysis of the task and performance
– Derivate: KLM

Both models rely on Cognitive System models (see previous chapters)

Benyon et al, Designing Interactive systems, chapter 20

Hierarchical Task Analysis (HTA)

Hierarchical Task Analysis (HTA)

graphical representation of a task structure

structure chart notation:

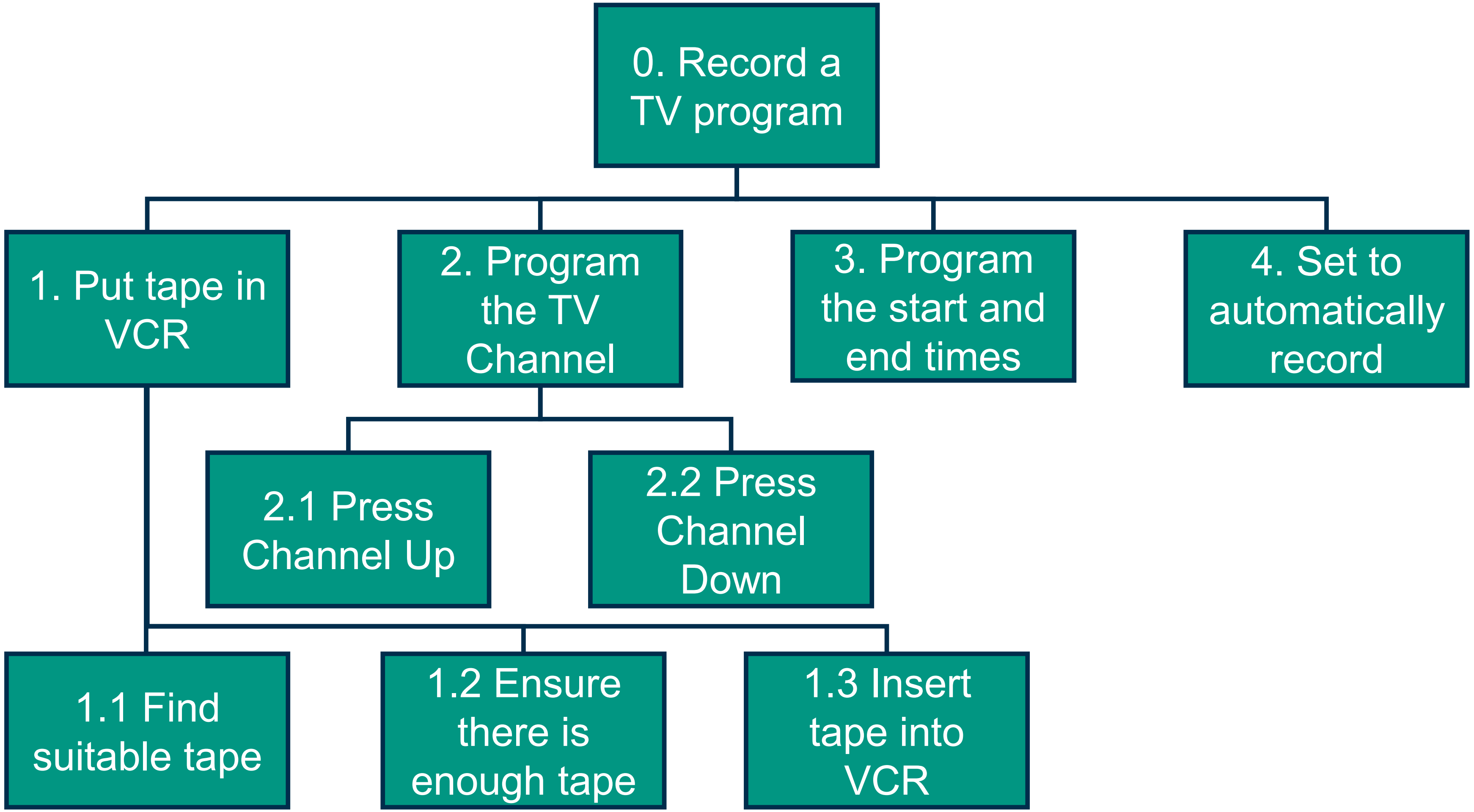
- sequence of tasks
- subtasks and actions as hierarchy
- actions can be repeated (iteration)
- alternative actions possible (selection)
- optional: annotations to indicate plans

not trivial:

- acquisition of task and subtask descriptions
- hierarchical modelling
- iterative process

Benyon et al, Designing Interactive systems, chapter 20

Hierarchical Task Analysis (HTA)



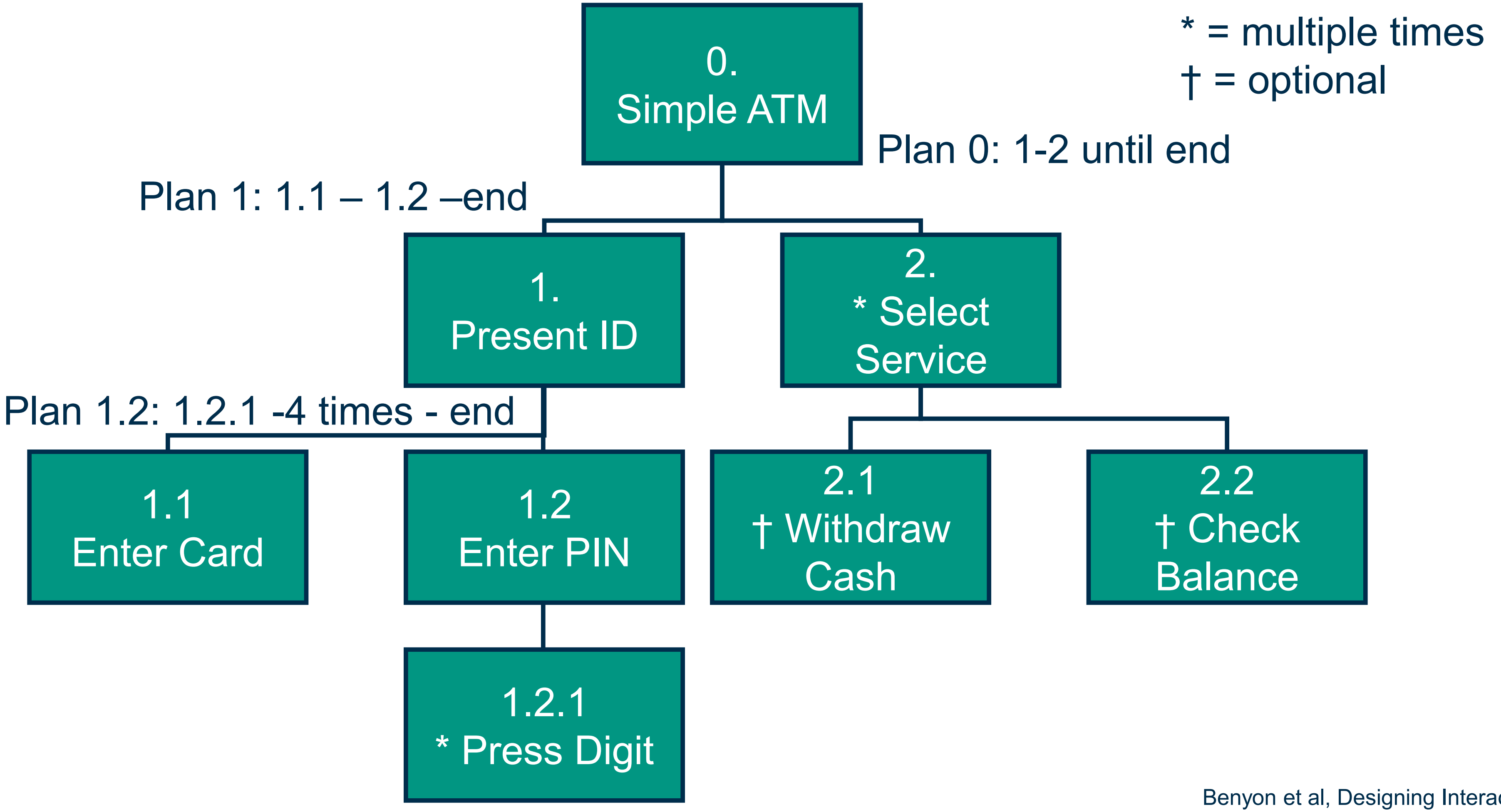
Hierarchical Task Analysis (HTA) – Step-by-Step

How to do an HTA step-by-step

- Decide on the purpose of the analysis
- Define the task goals
- Data acquisition: How to collect data?: people "logging", prototypes
- Acquire data and draft a hierarchical diagram
- Recheck validity of decomposition with stakeholders
- Identify significant operations and stop when effects of failure are no longer significant
- Generate and test hypotheses concerning factors affecting learning and performance

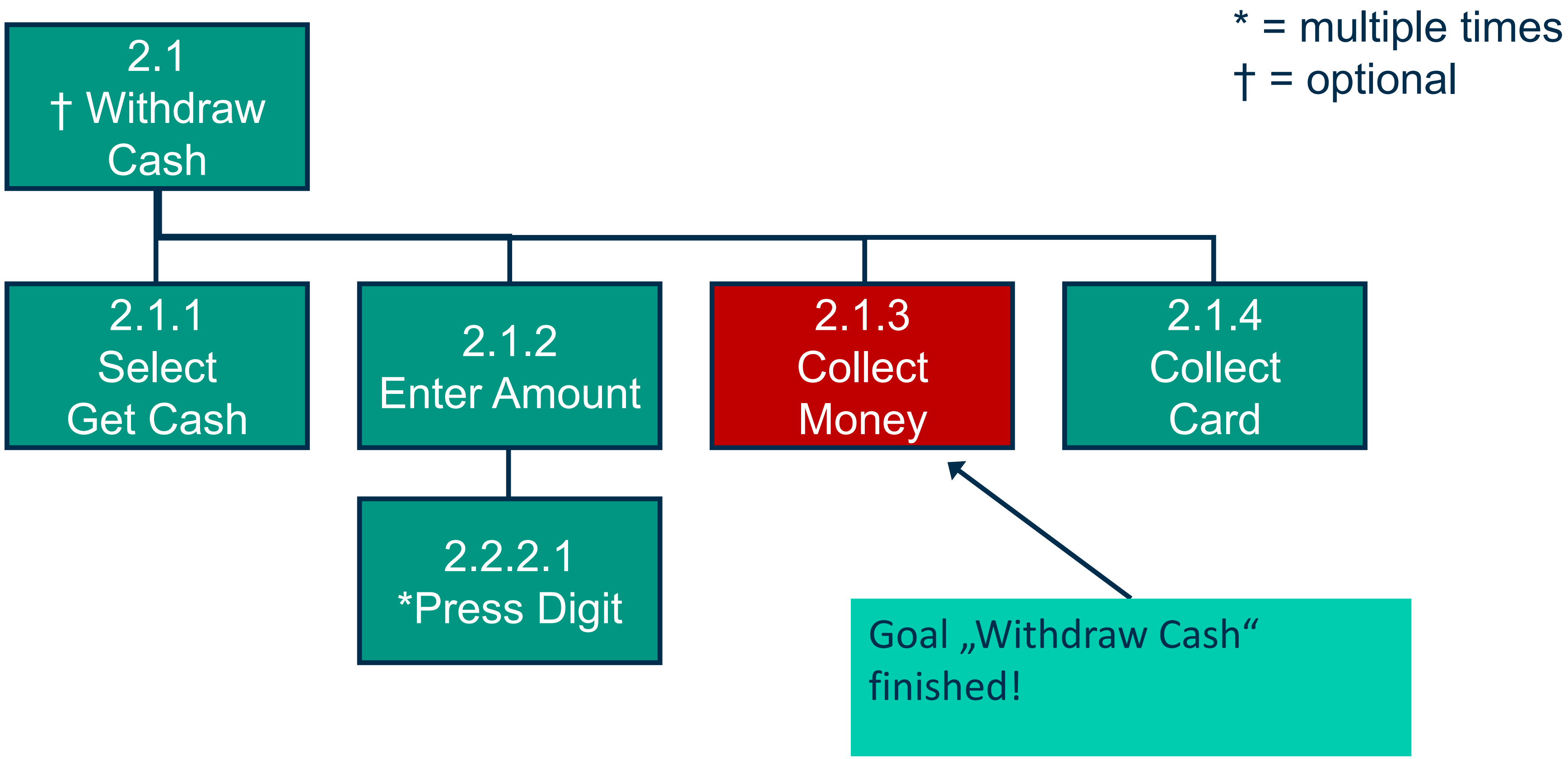
Annett, "Hierarchical task analysis", 2004
Benyon et al, Designing Interactive systems, chapter 20

HTA – Example: ATM adding options



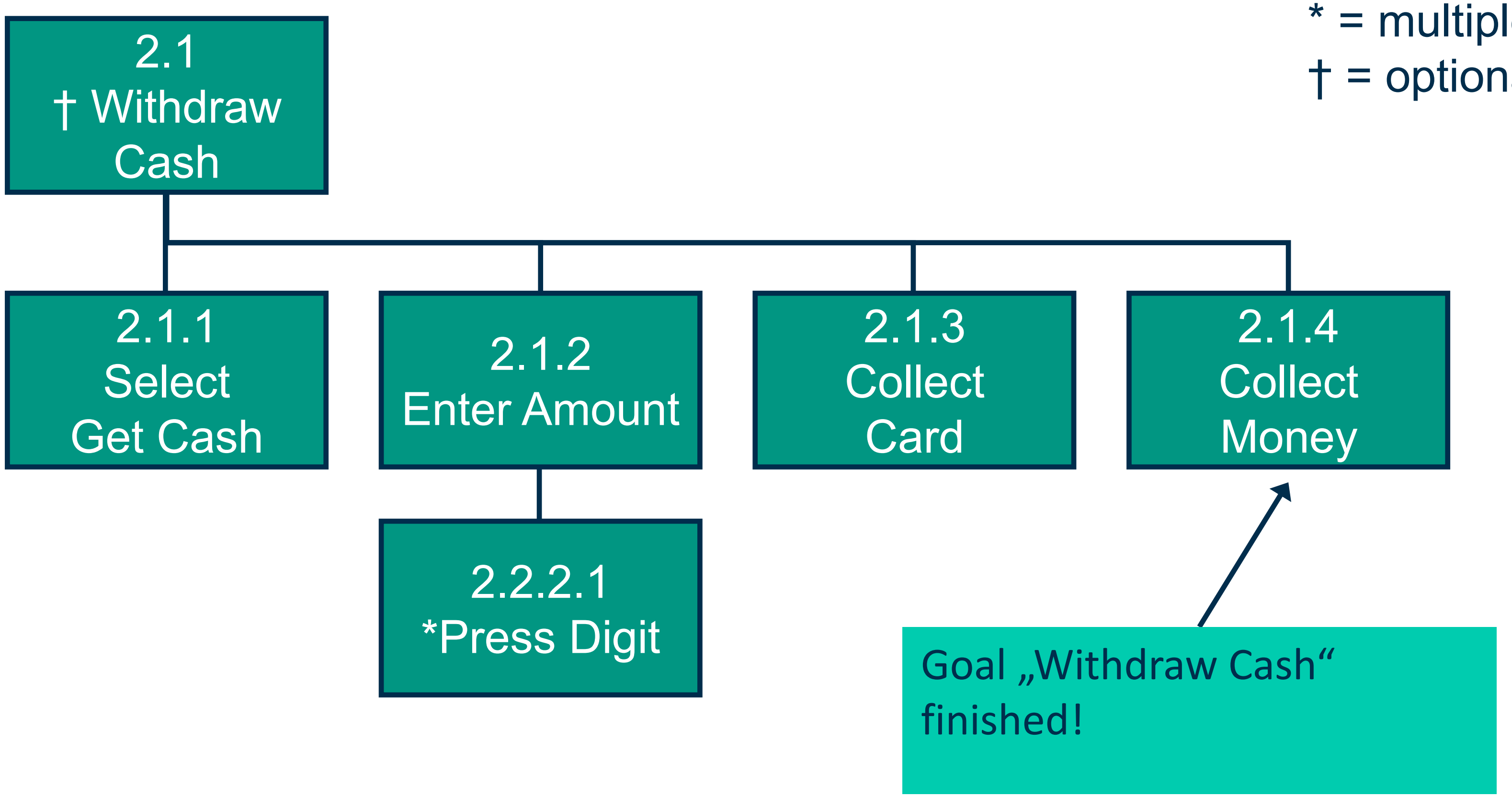
Benyon et al, Designing Interactive systems, chapter 20

HTA – Example: ATM adding options



Benyon et al, Designing Interactive systems, chapter 20

HTA – Example: ATM adding options



* = multiple times
† = optional

Benyon et al, Designing Interactive systems, chapter 20

Pingo

Question:
Answer question about “HTA”
Link:
https://pingo.coactum.de/589809

Analysis Tools →

goal-based task analysis (GOMS)

Original GOMS (CMN-GOMS)

Goals, Operators, Methods, Selection Rules

- An application of Human Information Processing and Task Analysis/Decomposition
- Focusing on Cognitive Load Analysis
- Introduced by Card, Moran & Newell (1983)

Explicit task structure

- hierarchy of goals and sub-goals
- Procedure: define goals and refine them
- Outcome are several sheets of GOMS descriptions, starting with the topmost description
- Methods in the topmost description are subsequently refined in the “lower” GOMS descriptions

GOMS

- John presented an example where a GOMS analysis showed that a new system would slow down work to factor 0.63
- The modelling part of the analysis had an effort of 2 MM + 18 MM field trials
- But saved 2 Million

CMN-GOMS

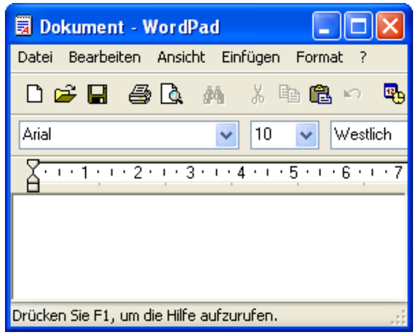
Engineering model of user interaction

- Goals - user's intentions (tasks)
  - e.g. write a text
  - Something the user perceives as a task
- Methods (Subgoals) - sequences of actions (operators)
  - may be multiple methods for accomplishing same goal
  - e.g. shortcut key or menu selection
- Operators - actions to complete task
  - Basic, low level action to perform or basic mental state
  - cognitive, perceptual & motor (HIP)
  - E.g. press X, read dialog
  - Time of activity can be given
- Selections - rules for choosing appropriate method
  - method predicted based on context
  - Selection rules are separately described

Close the window that has the focus (Windows XP)

Compare three options

ALT + F4



Key-shortcut

Context-menu

Close-button

GOAL: CLOSE-WINDOW
[select GOAL: USE-KEY-SHORTCUT
hold-ALT-key
press-F4-key
GOAL: USE-CONTEX-MENU
Move-mouse-win-head
Open-menu (right click)
Left-click-close
GOAL: USE-CLOSE-BUTTON
Move-mouse-button
Left-click-button]

Rule 1: USE-CLOSE-BUTTON method if no other rule is given
Rule 2: USE-KEY-SHORTCUT method if no mouse is present

Example (adapted from Dix 2004, p. 424): Copy a journal article

GOAL: PHOTOCOPY-PAPER
. GOAL: LOCATE-ARTICLE
. GOAL: COPY-PAGE repeat until no more pages
. GOAL: ORIENT-PAGE
. . OPEN-COVER
. . SELECT-PAGE
. . POSITION-PAGE
. . CLOSE-COVER
. GOAL: PRESS-COPY
. GOAL: VERIFY-COPY
. . LOCATE OUTPUT
. . EXAMINE COPY

. GOAL: COLLECT-COPY
. . LOCATE OUTPUT
. . REMOVE-COPY
. . (outer goal satisfied!)
. GOAL: RETRIEVE-ORIGINAL
. . OPEN-COVER
. . TAKE-ORIGINAL
. . CLOSE-COVER

Likely that the users forget this

Example (adapted from Dix 2004, p. 430): Cash-Machine: Why you need to get your card

Design to lose your card..

GOAL: GET-MONEY
. GOAL: USE-CASH-MACHINE
. . INSERT-CARD
. . ENTER-PIN
. . SELECT-GET-CASH
. . ENTER-AMOUNT
. . COLLECT-MONEY
. . (outer goal satisfied!)
. . COLLECT-CARD

Design to keep your card..

GOAL: GET-MONEY
. GOAL: USE-CASH-MACHINE
. . INSERT-CARD
. . ENTER-PIN
. . SELECT-GET-CASH
. . ENTER-AMOUNT
. . COLLECT-CARD
. . COLLECT-MONEY
. . (outer goal satisfied!)

GOMS – Other GOMS Models

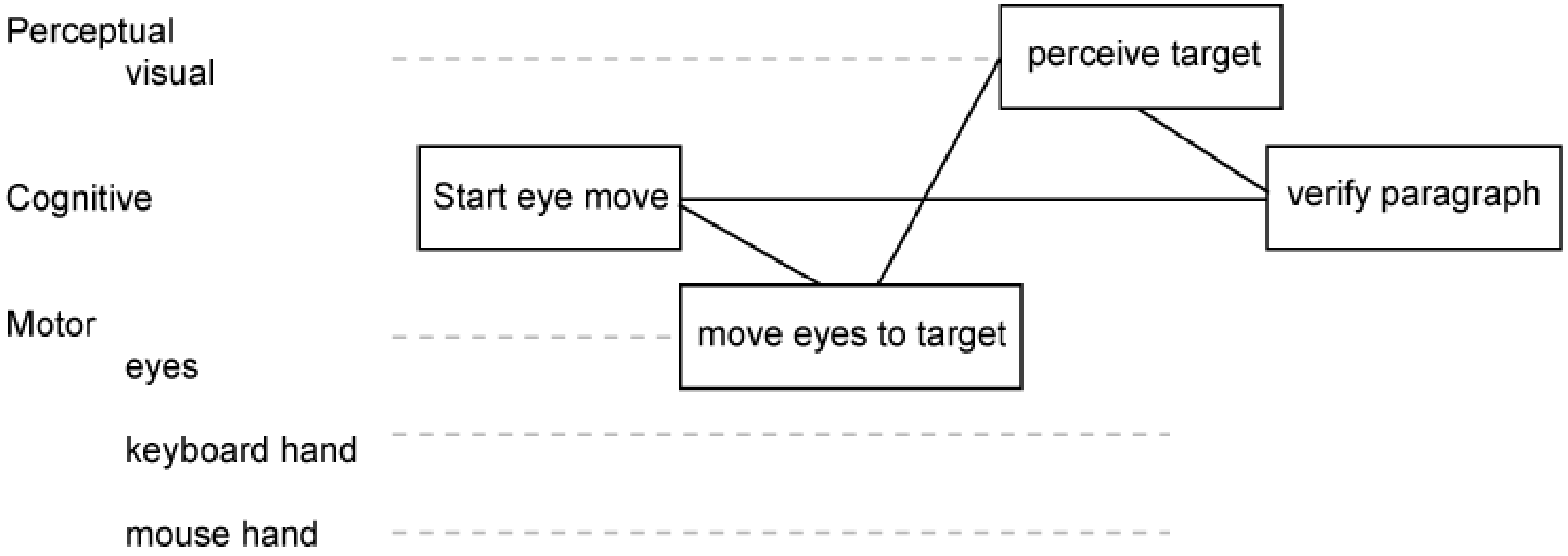
CPM-GOMS represents
- Cognitive
- Perceptual
- Motor operators

CPM-GOMS uses Program Evaluation Review Technique (PERT) charts
- Maps task durations using the critical path method (CPM).

CPM-GOMS is based directly on the Model Human Processor
- Assumes that perceptual, cognitive, and motor processors function in parallel

GOMS – Other GOMS Models

Program Evaluation Review Technique (PERT) chart Resource Flows



Example GOMS Example: Calculation interaction times

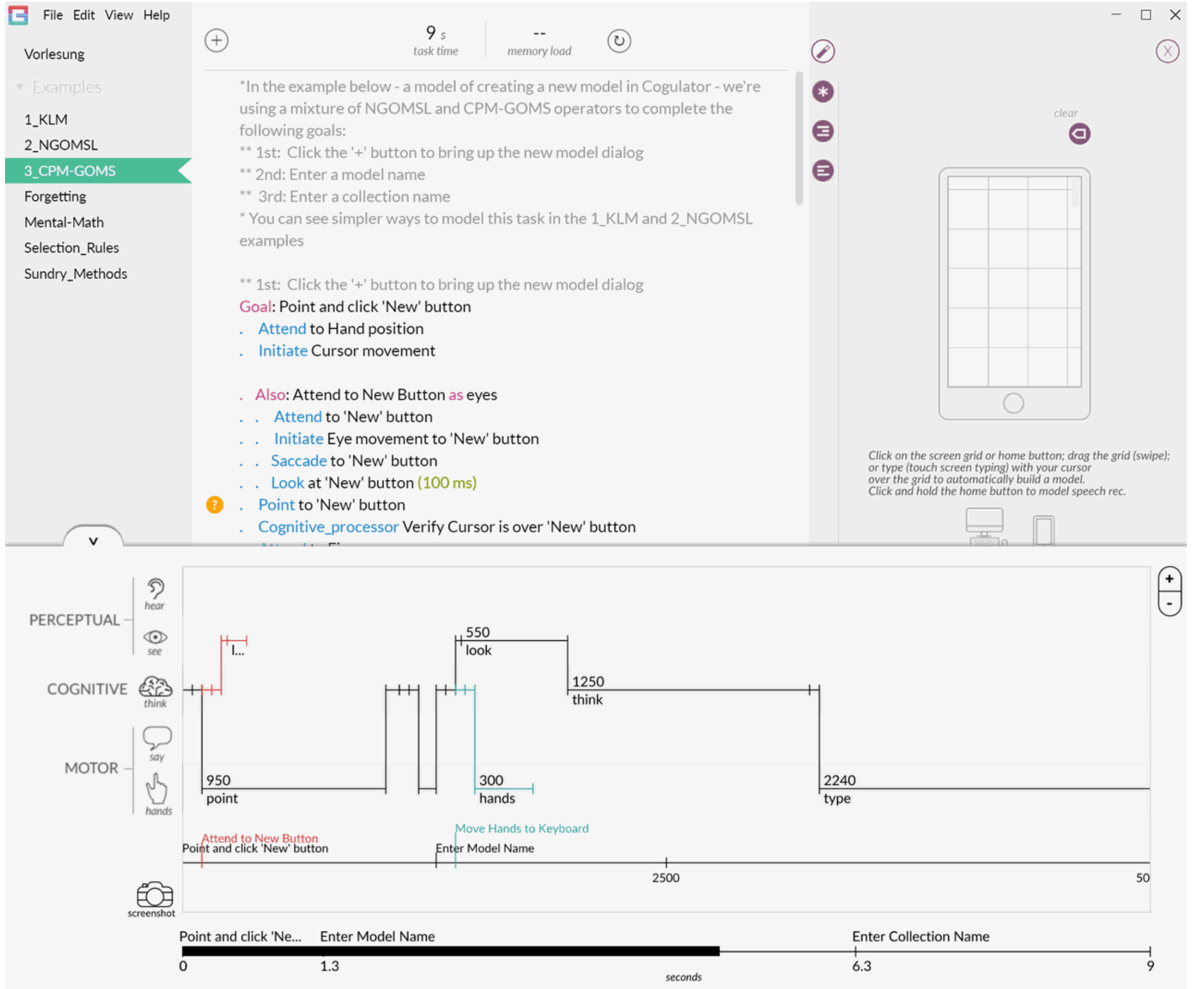
Method 1,2

Operators

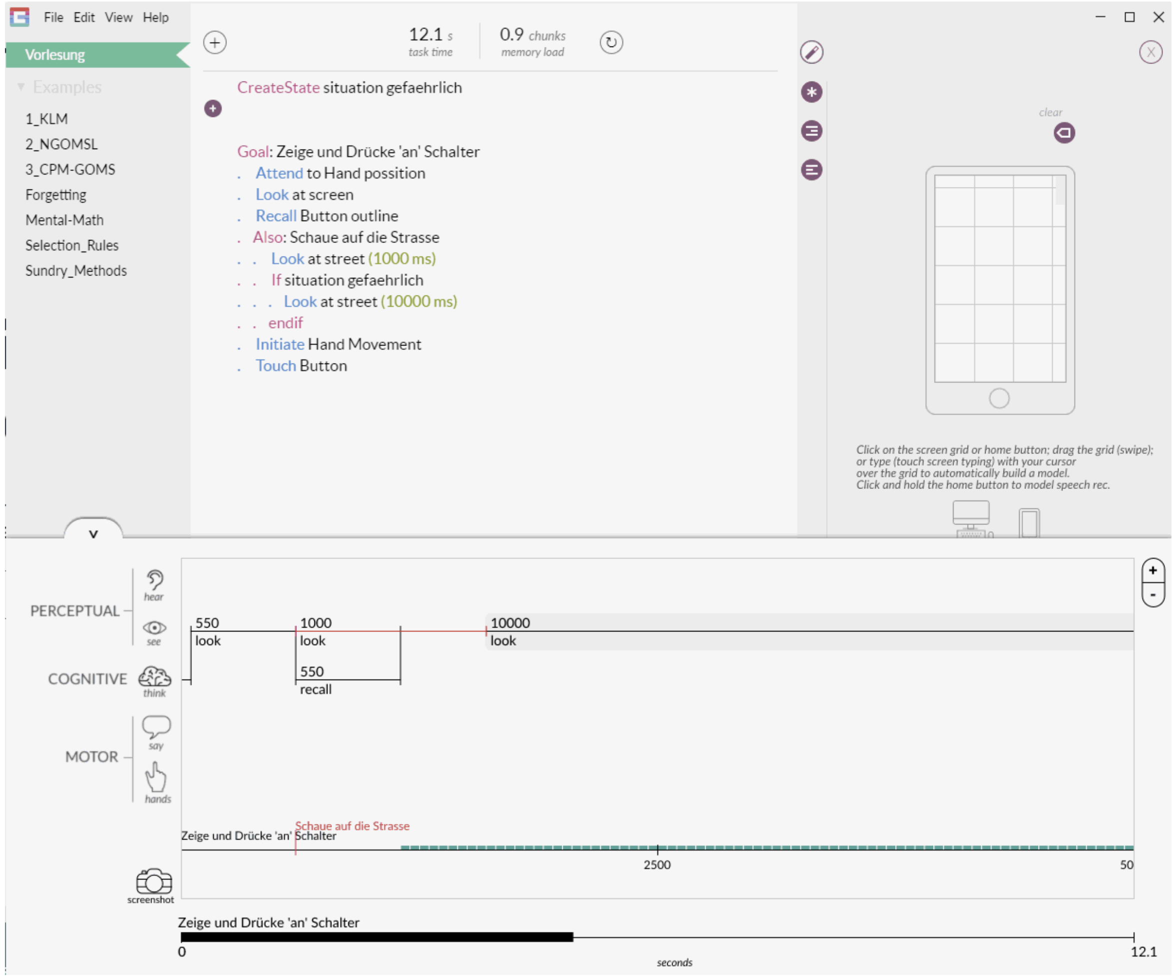
```
*Expansion of MOVE-TEXT goal
GOAL: MOVE-TEXT
.   GOAL: CUT-TEXT
.   .   GOAL: HIGHLIGHT-TEXT
.   .   .   [select**:  GOAL: HIGHLIGHT-WORD
.   .   .                   .   MOVE-CURSOR-TO-WORD
.   .   .                   .   DOUBLE-CLICK-MOUSE-BUTTON
.   .   .                   .   VERIFY-HIGHLIGHT
.   .   .               GOAL: HIGHLIGHT-ARBITRARY-TEXT
.   .   .                   .   MOVE-CURSOR-TO-BEGINNING            1.10
.   .   .                   .   CLICK-MOUSE-BUTTON                  0.20
.   .   .                   .   MOVE-CURSOR-TO-END                  1.10
.   .   .                   .   SHIFT-CLICK-MOUSE-BUTTON            0.48
.   .   .                   .   VERIFY-HIGHLIGHT]                   1.35
.   .   GOAL: ISSUE-CUT-COMMAND
.   .   .   MOVE-CURSOR-TO-EDIT-MENU1.10
.   .   .   PRESS-MOUSE-BUTTON                                      0.10
.   .   .   MOVE-MOUSE-TO-CUT-ITEM                                  1.10
.   .   .   VERIFY-HIGHLIGHT                                        1.35
.   .   .   RELEASE-MOUSE-BUTTON                                    0.10
.   GOAL: PASTE-TEXT
.   .   GOAL: POSITION-CURSOR-AT-INSERTION-POINT
.   .   .   MOVE-CURSOR-TO-INSERTION-POINT                          1.10
.   .   .   CLICK-MOUSE-BUTTON                                      0.20
.   .   .   VERIFY-POSITION                                         1.35
.   .   GOAL: ISSUE-PASTE-COMMAND
.   .   .   MOVE-CURSOR-TO-EDIT-MENU1.10
.   .   .   PRESS-MOUSE-BUTTON                                      0.10
.   .   .   MOVE-MOUSE-TO-PASTE-ITEM1.10
.   .   .   VERIFY-HIGHLIGHT                                        1.35
.   .   .   RELEASE-MOUSE-BUTTON                                    0.10
                                   TOTAL TIME PREDICTED (SEC)      14.38

**Selection Rule for GOAL: HIGHLIGHT-TEXT:
 If the text to be highlighted is a single word, use the
 HIGHLIGHT-WORD method, else use the HIGHLIGHT-ARBITRARY-TEXT method.
```

GOMS Open Source Tool

http://cogulator.io/



GOMS Open Source Tool



Keystroke Level Model (KLM)

Keystroke Level Model (KLM)

The KLM is a practical design tool that can capture and calculate the physical actions a user will have to carry out to complete specific tasks

The KLM can be used to determine the most efficient method and its suitability for specific contexts.

Given:

- A task (possibly involving several subtasks)
- The command language of a system
- The motor skill parameter of the user
- The response time parameters

Predict: The time an expert user will take to execute the task using the system

- Provided that he or she uses the method without error

Text-Editing Example variant: GOMS vs. Keystroke Level Model

```
GOAL: EDIT-MANUSCRIPT
.  GOAL: EDIT-UNIT-TASK ...repeat until no more unit tasks
.  .  GOAL: ACQUIRE UNIT-TASK ...if task not remembered
.  .  .  GOAL: TURN-PAGE ...if at end of manuscript page
.  .  .  GOAL: GET-FROM-MANUSCRIPT
.  .  GOAL:EXECUTE-UNIT-TASK ...if a unit task was found
.  .  .  GOAL: MODIFY-TEXT
.  .  .  .  [select: GOAL:  MOVE-TEXT* ...if text is to be moved
.  .  .  .             GOAL:  DELETE-PHRASE ...if a phrase is to be deleted
.  .  .  .             GOAL:  INSERT-WORD] ...if a word is to be inserted
.  .  .  .   VERIFY-EDIT

*Expansion of MOVE-TEXT goal
GOAL: MOVE-TEXT
.  GOAL: CUT-TEXT
.  .  GOAL: HIGHLIGHT-TEXT
.  .  .  [select**:  GOAL: HIGHLIGHT-WORD
.  .  .                 .  MOVE-CURSOR-TO-WORD
.  .  .                 .  DOUBLE-CLICK-MOUSE-BUTTON
.  .  .                 .  VERIFY-HIGHLIGHT
.  .  .              GOAL: HIGHLIGHT-ARBITRARY-TEXT
.  .  .                 .  MOVE-CURSOR-TO-BEGINNING                1.10
.  .  .                 .  CLICK-MOUSE-BUTTON                      0.20
.  .  .                 .  MOVE-CURSOR-TO-END                      1.10
.  .  .                 .  SHIFT-CLICK-MOUSE-BUTTON                0.48
.  .  .                 .  VERIFY-HIGHLIGHT]                       1.35
.  .  GOAL: ISSUE-CUT-COMMAND
.  .  .  MOVE-CURSOR-TO-EDIT-MENU1.10
.  .  .  PRESS-MOUSE-BUTTON                                        0.10
.  .  .  MOVE-MOUSE-TO-CUT-ITEM                                    1.10
.  .  .  VERIFY-HIGHLIGHT                                          1.35
.  .  .  RELEASE-MOUSE-BUTTON                                      0.10
.  GOAL: PASTE-TEXT
.  .  GOAL: POSITION-CURSOR-AT-INSERTION-POINT
.  .  .  MOVE-CURSOR-TO-INSERTION-POINT                            1.10
.  .  .  CLICK-MOUSE-BUTTON                                        0.20
.  .  .  VERIFY-POSITION                                           1.35
.  .  GOAL: ISSUE-PASTE-COMMAND
.  .  .  MOVE-CURSOR-TO-EDIT-MENU1.10
.  .  .  PRESS-MOUSE-BUTTON                                        0.10
.  .  .  MOVE-MOUSE-TO-PASTE-ITEM1.10
.  .  .  VERIFY-HIGHLIGHT                                          1.35
.  .  .  RELEASE-MOUSE-BUTTON                                      0.10
                                        TOTAL TIME PREDICTED (SEC)  14.38

 **Selection Rule for GOAL: HIGHLIGHT-TEXT:
  If the text to be highlighted is a single word, use the
  HIGHLIGHT-WORD method, else use the HIGHLIGHT-ARBITRARY-TEXT method.
```

Moving text with the *MENU-METHOD*

Description	Operator
Mentally prepare by Heuristic Rule 0	M
Move cursor to beginning of phrase	P
(no M by Heuristic Rule 1)	
Click mouse button	K
(no M by Heuristic Rule 0)	
Move cursor to end of phrase	P
(no M by Heuristic Rule 1)	
Shift-click mouse button	
(one average typing K)	K
(one mouse button click K)	K
Mentally prepare by Heuristic Rule 0	M
Move cursor to Edit menu	P
(no M by Heuristic Rule 1)	
Press mouse button	K
Move cursor to Cut menu item	P
(no M by Heuristic Rule 1)	
Release mouse button	K
Mentally prepare by Heuristic Rule 0	M
Move cursor to insertion point	P
Click mouse button	K
Mentally prepare by Heuristic Rule 0	M
Move cursor to Edit menu	P
(no M by Heuristic Rule 1)	
Press mouse button	K
Move cursor to Paste menu item	P
(no M by Heuristic Rule 1)	
Release mouse button	K

Keystroke Level Model (KLM)

The KLM is comprised of:
- Operators
- Encoding methods
- Heuristics for the placement of mental **(M)** operators

Operators
- **K** Press a key or button
- **P** Point with mouse
- **H** Home hands to keyboard or peripheral device
- **(D** Draw line segments)
- **M** Mental preparation
- **R** System response

KLM: Keystroke-Level Model

5 Basic by operations by Card, Moran and Newell:

K (0.2 s) – press a key or mouse button

P (1.1 s) – point with mouse

H (0.4 s) – home on keyboard, mouse or other device

M (1.35 s) – mentally prepare

R (t) – system response time
(needs to be measured)

http://www.syntagm.co.uk/design/klmcalc.shtml